Science Teacher, 5A

Vasant Valley School, New Delhi

Subject: Science

3139/S5A/0003

We are pleased to present the ASSET Teacher MyBook, a comprehensive resource designed to provide you with valuable insights into your students performance and support their academic growth. This resource offers a range of features to help you better understand your students' strengths and areas for development.

Personalized Teacher Feedback: Gain tailored insights into your students' distinctive strengths and areas for development. This feedback is aimed at assisting you in providing targeted guidance to each student.
Student Performance Table: Delve into the individual question-wise performances of your students. This data-driven view will aid you in identifying specific areas where students may require additional support.
Bird's Eye View: Acquire a holistic comparison of your students' performance against the national average. This perspective will help you contextualize your students' achievements and progress.
Year on Year Analysis - Previous Batch: This report compares student performance over a year.It compares the performance of a specific class compared to the previous year.
Skill-based Summary: Explore a tabulated display of students' skill proficiency, emphasizing the skills with the highest and lowest performance. This summary enables you to gauge skill performance relative to the school and national average.
Analysis of Easy/Difficult Questions: Identify questions posing ease or challenge compared to the national norm. This section illuminates areas needing focused attention, allowing you to tailor your teaching approach accordingly.
Questions with Common Wrong Answer: This section highlights potential misconceptions through selected questions. It provides valuable insights into your students' performance within a broader nationwide context.
Practice Questions and Explanation: In the Student MyBook, students receive practice questions tailored to their needs. This Teacher MyBook also features practice questions customized for your class, complete with answers and explanations. Different practice approaches have been thoughtfully curated based on subject and skill requirements.

We recommend that you review this analysis with reference to the ASSET question paper. Your commitment to your students' growth is highly commendable, and we believe that the ASSET Teacher MyBook will further empower you in this journey.

Should you have any questions or require further assistance, please feel free to reach out to us at info@ei.study. We are dedicated to supporting your efforts in enhancing your students' learning experience.

Regards,
The ASSET Team

Kindly note: Some images may slightly appear unclear due to technical reasons, but rest assured, this is not an error.

Dear Science Teacher,

Congratulations on receiving your students' ASSET test results!

ASSET helps us understand where your students excel and where they might need more support.

Let's celebrate the strengths your students have shown in Science are:

Recollection or recognition of science facts and concepts
Representing, relating or explaining scientific processes or observed phenomena

Great job nurturing these talents!

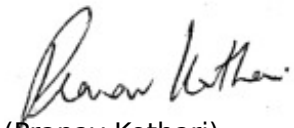
Now, let's look at areas where we can help them grow even more. Here's where they can improve in Science :

Extraction, translation and application of knowledge or information
Knowledge of use of scientific instruments, tools and procedures

To assist you in this journey, we've created a special booklet. It's designed to focus on the questions/ skills that need a little boost. The booklet provides insights into each student's progress and highlights questions that sparked interest or posed a challenge.

Take your time going through the analysis to understand how each student is doing. You'll find different sections in the booklet to spot learning gaps and dive into specific questions that need attention. Your students have personalized booklets, called Student MyBooks, too. These are tailored to their needs, giving them a glimpse into the areas where they can practice more.

Feel free to use these resources in class, and let us know how it goes. We're here to help and learn from your experience. Share your thoughts with us at info@ei.study.As you guide your students through their learning journey, you're helping them become even better. Best of luck!



(Pranav Kothari)
Chief Executive Officer - Educational Initiatives

In case of queries, please write to us at info@ei.study or contact our customer care executive at the toll free number 1800-102-8885 (9 am - 6 pm, Mon - Fri).

No.	Student↓	Question No: → / Correct Answers →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	Score	Scaled Scores	Percentile	Performance
			D	D	A	A	B	D	C	B	A	C	B	D	D	C	D	A	B	D	A	B	A	B	D	D	B	A	A	B	C	D	A	B	B	D	B				
1	Adhrit P Chandra		✓	✓	✓	✓	✓	✓	✓	✓	C	✓	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	✓	✓	C	✓	✓	✓	D	✓	✓	✓	✓	✓	✓	✓	✓	31	800	99	O
2	Akira Nagpal		✓	✓	C	B	✓	C	✓	✓	✓	✓	✓	✓	✓	✓	B	✓	✓	B	✓	✓	✓	✓	C	✓	✓	✓	✓	C	✓	✓	B	A	✓	✓	A	25	670	94	
3	Annika I Dayal		✓	✓	✓	B	✓	✓	A	D	D	✓	✓	✓		D	B	B	✓	B	D	A	✓	A	✓	C	A	✓	✓	C	✓	✓	✓	A	A	✓	C	17	540	61	
4	Aranya D Bedi		A	A	C	B	D	C	B	✓	B	A	✓	✓	✓	B	A	✓	✓	*	✓	✓	✓	✓	✓	*	✓	B	✓	✓	D	✓	✓	✓	✓	C	✓	19	569	71	
5	Arav Saha		B	B	C	D	C	✓	D	C	B	✓	✓	✓	✓	B	A	✓	✓	✓	B	✓	✓	A	A	✓	✓	B	✓	C	✓	A	C	✓	C	✓	✓	17	557	67	
6	Archisha Singh		✓	✓	C	B	C	C	✓	✓	C	✓	✓	C	✓	✓	✓	B	A	B	D	✓	✓	C	B	C	✓	✓	B	A	✓	A	✓	✓	✓	✓	A	18	537	60	
7	Asta A Dayal		C	✓	C	C	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	B	B	✓	A	✓	C	D	A	✓	✓	✓	✓	D	✓	✓	✓	✓	✓	C	C	A	22	610	82	
8	Ayaan Malik		C	✓	D	C	D	B	A	D	C	D	D	✓	✓	B	B	C	✓	A	C	A	✓	C	C	✓	C	D	B	✓	B	B	✓	C	C	C	D	8	402	14	
9	Ayaan Tandon		B	A	*	✓	*	✓	A	D	B	✓	D	*	✓	✓	✓	C	✓	A	✓	D	✓	A	✓	✓	✓	B	✓	A	*	✓	B	✓	✓	B	C	16	507	49	
10	Ayat Rehman		A	A	C	B	✓	B	D	D	B	A	✓	B	✓	✓	A	C	✓	A	D	✓	✓	A	✓	B	✓	✓	B	✓	✓	✓	✓	✓	D	✓	A	16	525	55	
11	Bharti		A	A	C	B	D	✓	✓	D	D	✓	A	A	✓	D	A	✓	✓	C	B	✓	C	A	C	C	D	✓	D	✓	✓	A	B	A	C	A	A	10	429	22	
12	Dipanshi		✓	✓	C	B	D	✓	✓	✓	✓	✓	D	✓	A	B	B	C	✓	✓	✓	A	B	D	C	✓	✓	✓	✓	C	✓	A	B	✓	✓	✓	A	19	561	68	
13	Gaia Bhagwani		B	✓	✓	B	C	C	✓	✓	B	✓	A	✓	✓	✓	✓	B	✓	✓	✓	D	✓	A	C	✓	✓	✓	B	✓	✓	C	✓	✓	C	✓	✓	22	614	84	
14	Karan Puri		✓	✓	C	D	✓	✓		C	C	✓					A	C								✓									✓	✓	✓	9	392	12	
15	Mohammad Faizan		C	A	✓	B	✓	✓	D		B	✓	A	B	✓	*	B	C	✓	A	C	A	*	*	C	C	D	✓	✓	C	A	B	D	✓	✓	✓	D	11	435	24	
16	Nabiya Chouhan		B	A	D	D	✓	C	B	A	B	D	C	A	✓	B	A	B	C	B	D	D	D	✓	A	✓	C	B	D	A	✓	B	D	✓	C	A	C	6	366	6	
17	Neel R Khanna		C	A	✓	D	✓	✓	✓	A	B	D	D	B	✓	✓	✓	B	A	B	C	A	✓	A	✓	✓	✓	D	D	✓	✓	✓	✓	✓	✓	✓	D	18	543	62	
18	Parth H Yadav		✓	✓	B	B	✓	✓	A	✓	C	✓	C	✓	✓	D	✓	B	✓	✓	D	✓	✓	✓	✓	✓	C	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	25	696	97	
19	Prathit Goyal		✓	A	✓	✓	D	B	B	D	C	A	A	B	✓	✓	C	B	✓	*	B	✓	✓	✓	✓	✓	✓	C	C	✓	✓	B	✓		✓		*	16	488	42	
20	Reyansh Chopra		A	A	B	B	C	B	B	D	C	D	✓	B		✓	A	C	✓	✓	✓	✓	✓	A	✓	C	D	✓	B	✓	✓	B	B	✓	C	C	✓	13	462	33	
21	Ruhani Tulshan		✓	A	D	D	✓	C	✓	✓	D	✓	✓	✓	✓	✓	✓	C	✓	✓	✓	✓	✓	✓	B	A	✓	C	D	✓	✓	✓	✓	✓	✓	✓	C	24	685	96	
22	Sameerudin Ansari		C	A	✓	✓	D	B	A	✓	C	✓	✓	✓	✓		A	✓	✓	A	D	C	B	C	✓	B	A	B	D	D	✓	A	✓	✓	✓	✓	A	15	481	39	
23	Samerah J Srinivas		✓	A	✓	✓	✓	✓	A	✓	✓	A	✓	✓	✓	D	✓	C	✓	✓	D	C	✓	✓	C	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	A	26	661	93	
24	Sanshray J Pratyush		✓	✓	✓	B	✓	C	A	C	✓	✓	✓	✓	✓	D	✓	C	✓	A	✓	✓	✓	✓	✓	A	✓	✓	✓	C	✓	✓	✓	✓	✓	✓	✓	26	687	96	
25	Shivai Dang		B	✓	C	B	C	B	D	D	✓	✓	D	✓	✓	✓	C	B	✓	A	✓	✓	✓	✓	B	✓	✓	✓	✓	A	✓	✓	B	✓	✓	*	✓	20	585	76	
26	Shreya D Bahree		✓	✓	✓	B	✓	C	A	D	D	D	✓	✓	✓	D	✓	B	✓	C	D	D	C	D	B	C	✓	✓	✓	✓	D	✓	✓	✓	C	✓	C	17	537	60	
27	Sofia C Sanjanwala		B	A	✓	B	C	C	A	D	C	✓	✓	✓	✓	✓	B	C	✓	✓	✓	D	✓	✓	C	B	✓	✓	✓	✓	D	✓	C	✓	✓	✓	D	19	580	74	
28	Taahira Trehan		✓	C	✓	B	✓	B	D	✓	✓	✓	✓	✓	✓	D	✓	C	✓	A	D	A	✓	D	✓	✓	D	B	D	✓	✓	✓	✓	✓	C	C	A	19	555	66	
29	Vaidehi Chaudhry		✓	✓	✓	✓	✓	✓	D	✓	B	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	A	A	B	✓	✓	✓	✓	✓	✓	C	✓	A	28	727	98	
30	Vieara Chaudhry		✓	✓	B	B	✓	✓	B	✓	✓	✓	✓	C	✓	✓	C	B	A	✓	D	✓	C	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	D	✓	A	C	22	578	74	

% answered correct (section)	50	53	44	21	62	44	32	44	29	71	65	65	85	47	38	21	82	35	47	50	74	50	50	56	65	65	50	56	79	62	65	76	59	65	32	566.4/500		
% answered correct (school students)	58	50	39	12	52	47	43	56	27	67	67	77	89	64	39	27	80	40	49	47	71	53	49	61	55	63	47	59	77	69	60	70	57	61	29	573.3/500		
% answered correct (all students)	44	34	35	21	42	51	60	58	21	51	55	65	71	52	35	27	70	47	38	34	57	45	40	56	42	47	42	52	56	47	55	58	42	55	26	508.7/500		

Notes: Score: Actual score obtained by the student. **Percentile:** Percentage of students who have scored less than that particular student. **Performance:**'O' indicates outstanding performance in subject.
✓ Indicates correct answer. **A,B,C,D:** Indicates student's incorrect answer. **(Blank)** indicates student omitted answer. ***** Indicates student chose more than one option.
Scaled Score: Raw scores are converted to a scaled score through a process known as equating. The minimum a student can get is 200 and the maximum is 800.

No.	Student↓	Question No: →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	Score	Scaled Scores	Percentile	Performance
		Correct Answers →	**D**	**D**	**A**	**A**	**B**	**D**	**C**	**B**	**A**	**C**	**B**	**D**	**D**	**C**	**D**	**A**	**B**	**D**	**A**	**B**	**A**	**B**	**D**	**D**	**B**	**A**	**A**	**B**	**C**	**D**	**A**	**B**	**B**	**D**	**B**				
31	Yadav Anushka		✓	✓	✓	B	✓	C	A	C	✓	✓	✓	✓	✓	D	✓	C	✓	A	✓	✓	✓	✓	✓	A	✓	✓	✓	C	✓	✓	✓	✓	✓	✓	✓	26	687	96	
32	Yash Yadav		B	A	✓	✓	✓	B	✓	D	B	A	✓	C	A	A	A	B	✓	A	B	D	✓	✓	C	✓	✓	✓	✓	A	✓	C	✓	✓	C	A	C	15	481	39	
33	Yuvaan Bhutani		✓	✓	C	B	✓	C	✓	✓	✓	✓	✓	✓	✓	✓	C	✓	D	✓	✓	A	✓	✓	✓	C	✓	✓	D	✓	✓	✓	✓	D	✓	✓	C	25	652	91	
34	Zayn Singh		A	✓	D	D	✓	✓	D	D	D	✓	✓	✓	✓	D	C	C	✓	B	✓	✓	✓	✓	✓	✓	D	✓	✓	C	✓	✓	✓	✓	D	✓	✓	22	659	92	

% answered correct (section)	50	53	44	21	62	44	32	44	29	71	65	65	85	47	38	21	82	35	47	50	74	50	50	56	65	65	50	56	79	62	65	76	59	65	32	566.4/500			
% answered correct (school students)	58	50	39	12	52	47	43	56	27	67	67	77	89	64	39	27	80	40	49	47	71	53	49	61	55	63	47	59	77	69	60	70	57	61	29	573.3/500			
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Notes: Score: Actual score obtained by the student. **Percentile:** Percentage of students who have scored less than that particular student. **Performance:**'O' indicates outstanding performance in subject.
✓ Indicates correct answer. **A,B,C,D:** Indicates student's incorrect answer. **(Blank)** indicates student omitted answer. * Indicates student chose more than one option.
Scaled Score: Raw scores are converted to a scaled score through a process known as equating. The minimum a student can get is 200 and the maximum is 800.

The Bird's Eye View shows a comparison of the performance of your students with the national average performance.

Your class 5 A, has performed better than the national average in the Science ASSET paper this year.

Class 5	Science	No. of students
5A (Your Class)	113%	34
5B	115%	32
5C	116%	32
All sections combined	115%	98

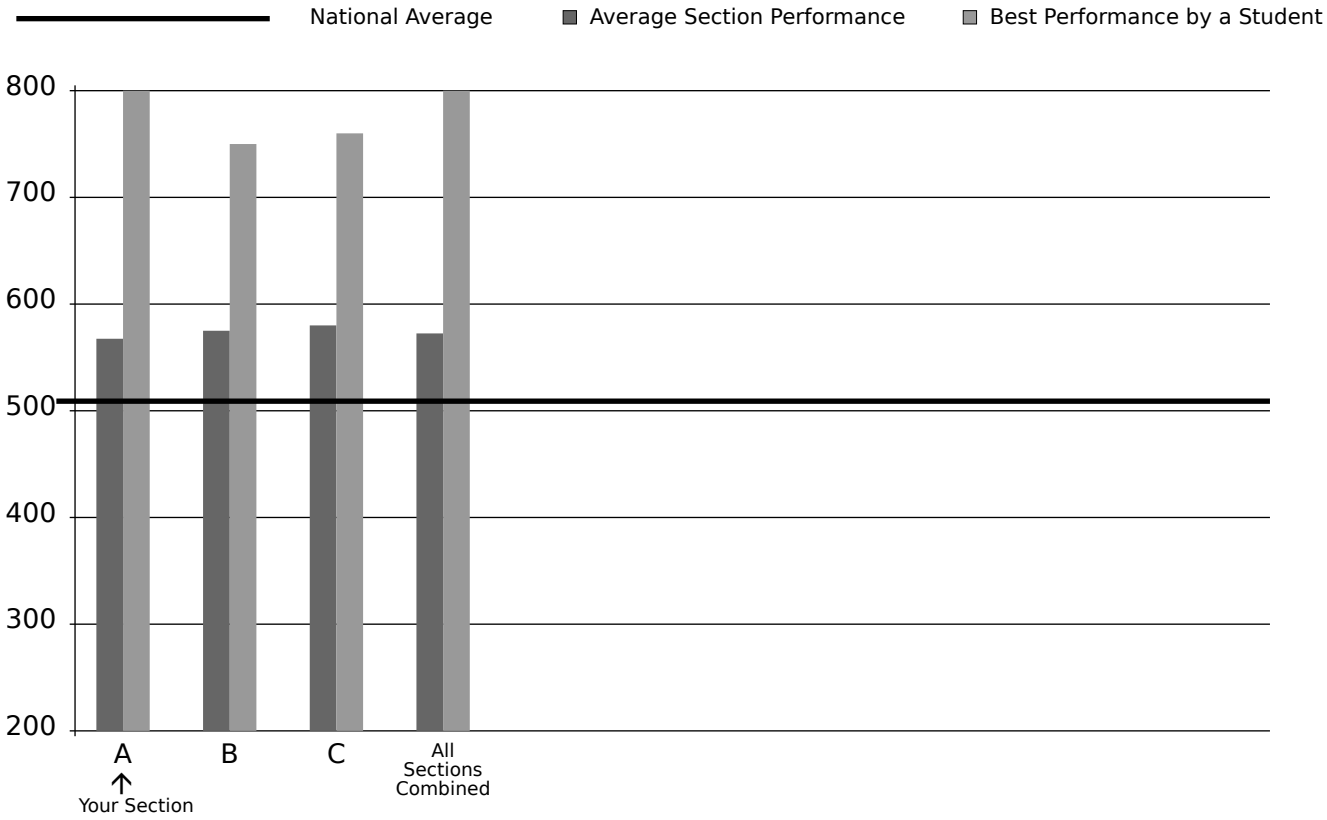
Performance as a percentage of National Average

Above Average(>105%) Below Average(<95%) Within Average(95% to 105%)

National average is taken as 100%. The value in the table indicate performance as compared to the national average.**E.g.** For a class average of 520 and overall average as 500, overall benchmarking %= (520/500)*100 = 104%. 95% to 105% is treated as similar to national average performance.Performance greater than 105% and below 95% is considered as performance above and below the national average range respectively.

Performance Data

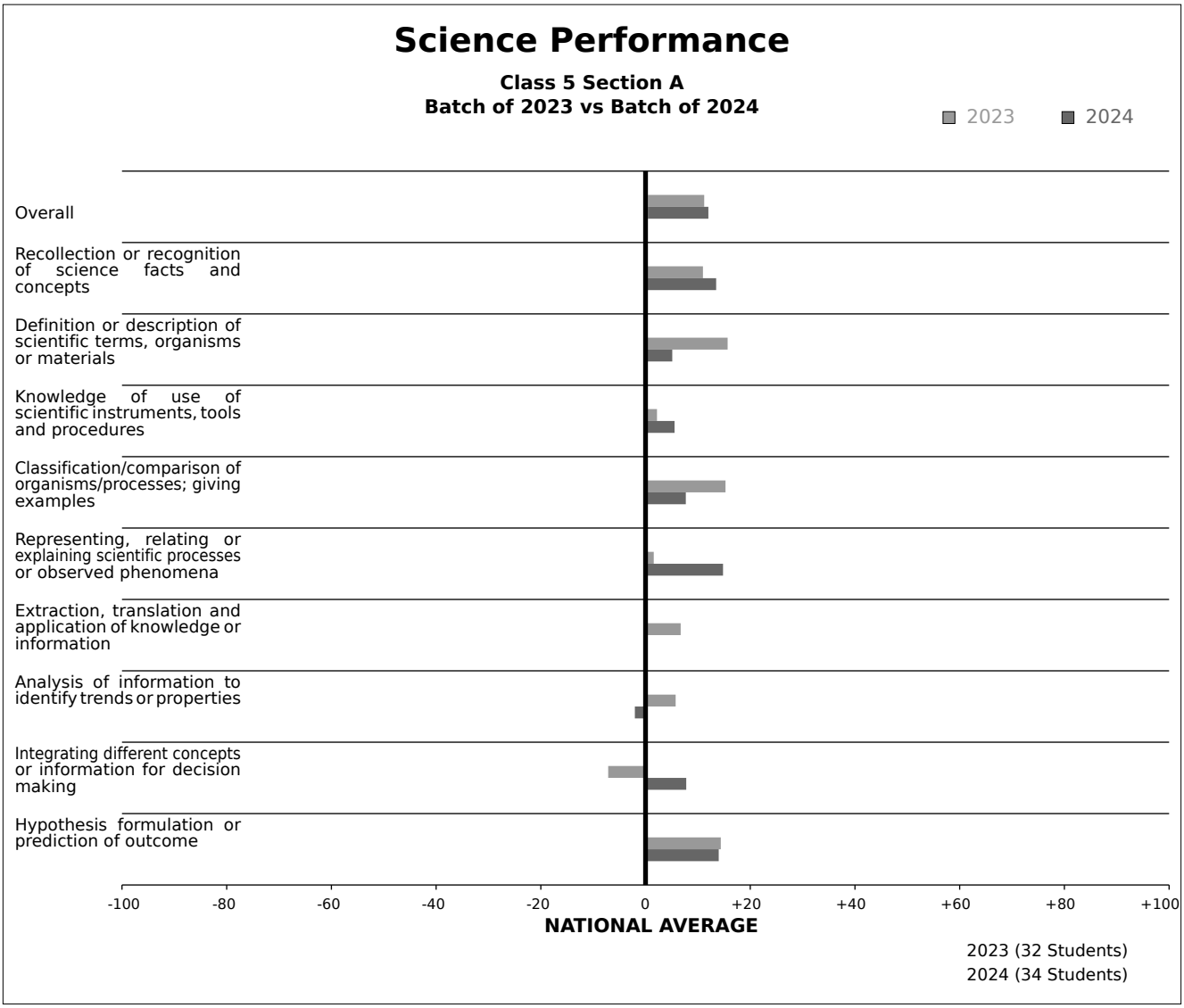
Comparison of section average and best performance with all schools in the country.



Comparison of performance of students of the current batch with the previous batch:

The following graph compares the performance of your current batch of students against the previous batch. The horizontal bars in the graph show how students of your class have fared against the national average in each skill across two years.

The thick black vertical line in the centre marks the national average.For example, a bar towards the right would mean that students performed higher than the national average that year in the particular skill.



This table shows how your students' performance compares to the overall school performance and the national average (all other schools in India). The skills with a '*' in the first column, if any, are those in which your students have performed better than the national average.

Class: 5 A Subject: Science Number of Students: 34

S.No.	Skill	Questions	Class 5 A Performance	Class 5 Performance (School)	Class 5 Performance (National)
			Average	Average	Average
1*	Recollection or recognition of science facts and concepts	13,17,26,34	74.3%	73.2%	60.8%
2*	Definition or description of scientific terms, organisms or materials	2,8,22,31	52.9%	54.8%	47.8%
3*	Knowledge of use of scientific instruments, tools and procedures	1,28,35	46.1%	48.6%	40.5%
4*	Classification/comparison of organisms/ processes; giving examples	5,12,16,19,21	53.5%	55.1%	45.8%
5*	Representing, relating or explaining scientific processes or observed phenomena	4,10,29,33	57.4%	53.3%	42.6%
6	Extraction, translation and application of knowledge or information	7,15,20,27	42.6%	43.9%	42.7%
7	Analysis of information to identify trends or properties	6,11,18,24	50.0%	53.8%	52.0%
8*	Integrating different concepts or information for decision making	3,14,32	55.9%	57.8%	48.1%
9*	Hypothesis formulation or prediction of outcome	9,23,25,30	51.5%	50.0%	37.5%

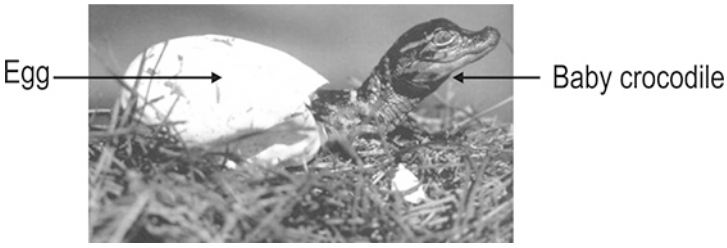
Note - The average performance is calculated from the number of correct responses given by students to the questions constituting specific skills.

Strong skills		Weak skills	
1.	Recollection or recognition of science facts and concepts	1.	Extraction, translation and application of knowledge or information
2.	Representing, relating or explaining scientific processes or observed	2.	Knowledge of use of scientific instruments, tools and procedures

This part of the report shows questions that the children of your class found easy and difficult. This information can be used by you to focus your attention to specific questions and skills. You can also use the student performance table to focus on individual students. (Please refer to the ASSET paper for the complete text of these questions.)

Questions which students of your school found the MOST DIFFICULT compared to the rest of the country:

Q 7 Here is a picture of a baby crocodile hatching from its egg.



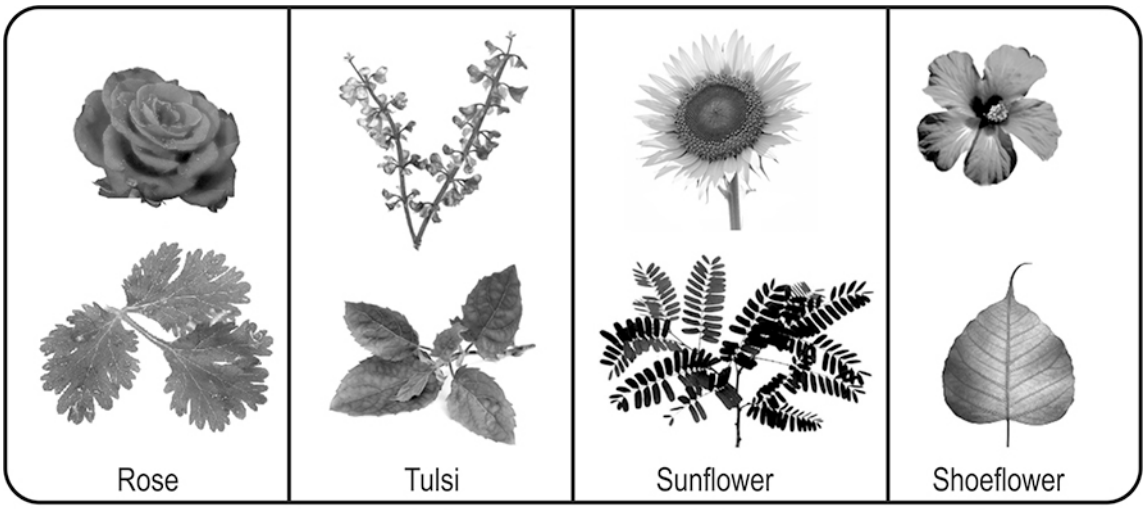
From where does the baby crocodile get food while it is growing inside the egg?

A A baby crocodile does not need food.
B The mother feeds the baby crocodile with milk.
C The egg has food stored for the baby crocodile.
D The baby crocodile can make its own food.

Skill: Extraction, translation and application of knowledge or information

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
7	35224	C	32%	60%	Your School	
					All Schools	

Q 8 Which flower is correctly matched with its leaves?



A **B** **C** **D**

Skill: Definition or description of scientific terms, organisms or materials

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
8	35224	B	44%	58%	Your School	
					All Schools	

Q 18 Diya wound a thick cotton thread around a wooden block as shown in Figure 1. She painted the block and the thread on it as shown in Figure 2.

Figure 1

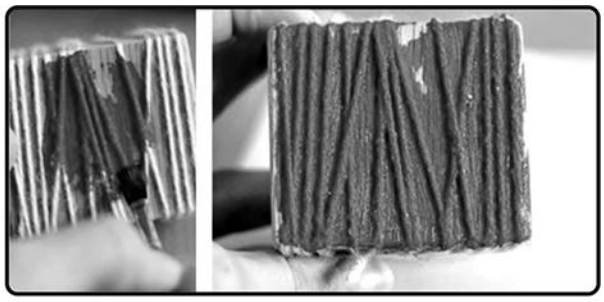


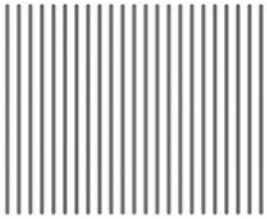
Figure 2

She GENTLY pressed the freshly painted block onto a white paper.

What will be printed on the paper?

A

B



C

D

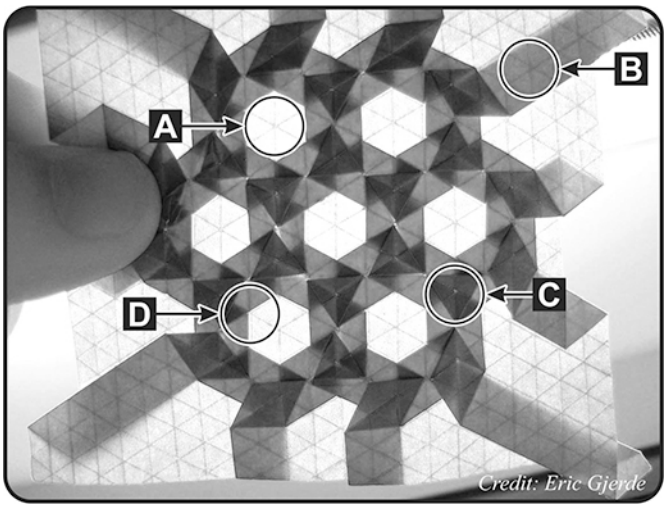
Skill: Analysis of information to identify trends or properties

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
18	35224	D	35%	47%	Your School	
					All Schools	

Questions which students of your school found the EASIEST compared to the rest of the country:

Q 29 Eric makes wonderful origami designs just by FOLDING a single piece of paper. The picture below shows one of his designs held in front of a light bulb.

Which of the marked parts has the MOST number of folds?



Skill: Representing, relating or explaining scientific processes or observed phenomena

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph
29	35224	C	79%	56%	Your School / All Schools

Q 25 Debbie made a list of all animals with hair on their body. She also noted the number of legs that each animal had.

What was she trying to find out from this list?

A if some animals with two legs have feathers on their body
B if some animals with hair on their body have less than four legs
C if there are more animals with hair on their body than animals with feathers
D if animals with hair on their body and animals with feathers on their body are equal in number

Skill: Hypothesis formulation or prediction of outcome

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph
25	35224	B	65%	42%	Your School / All Schools

Q 5 Simon was making a list of things that are man-made. Rama was making a list of things that sink in water.

Which of the following can be put in both the lists?

A feather **B** glass rod **C** wooden ball **D** thermocol sheet

Skill: Classification/comparison of organisms/processes; giving examples

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
5	35224	B	62%	42%	Your School	
					All Schools	

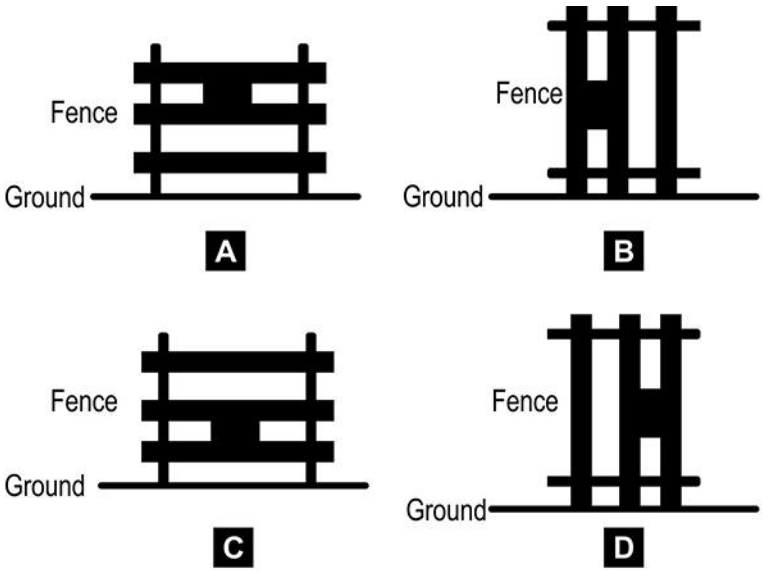
From the questions that students across the country have found the most difficult, we have chosen a few to highlight potential misconceptions that students might have. For each question, we have given a break-up (in percentage form) of the number of students choosing a particular option. This has been done both for the section of your school and all the students nationally. Compare these to find out how your students have performed.

We have highlighted in light grey the correct option and in dark grey the most common wrong answer, and provided an explanation of why students are choosing the most common wrong option. It is recommended that you discuss these questions with the students to uncover students' rationale in choosing the most common wrong option. (Please refer to the ASSET paper for the complete text of these questions.)

Q 4 The picture below shows the shadow of a fence on the ground.

Shadow of fence on the ground

Which of the following could be the fence?



The question tests students' ability TO relate an object WITH the shadow that it forms. Students choosing the correct OPTION A understand that the light falling ON the fence will CAST a shadow IN the same orientation ON the ground. Students choosing OPTION B might be influenced BY the similar orientation of the shadow AND the fence AND have NOT considered the orientation of the ground. Students choosing OPTION C might be confused about the orientation of the shadow WITH respect TO the object. Students choosing OPTION D might be thinking that the shadow will be reversed AND have also NOT considered the orientation of the ground.

Papercode: 35224, Question no: 4					
Performance	Option A	Option B	Option C	Option D	No. of Students
Section	21%	56%	6%	18%	34
National	21%	61%	6%	11%	13500

Q 3 Water is filled to the same level in the jars X and Y as shown in Figure 1. Pebble P is dropped into Jar X and Pebble Q is dropped into Jar Y. The water rises to the same level in the two jars as shown in Figure 2.

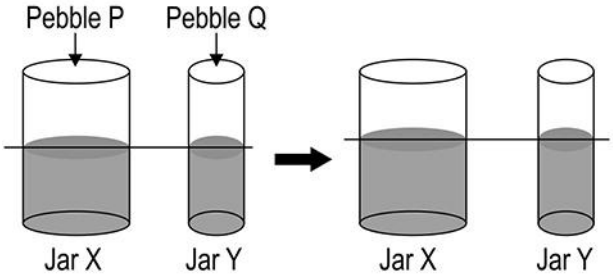


Fig. 1 **Fig. 2**

How does the size of the pebbles P and Q compare?

A Pebble P is bigger than Pebble Q.
B Pebble Q is bigger than Pebble P.
C The size of Pebble P is equal to the size of Pebble Q.
D (Cannot say as jars X and Y are not of the same size.)

The question tests students' understanding of relative volumes. Students choosing the correct option A understand that since Jar X is wider, more water needs to be displaced for the level of water to be raised to the same level as in Jar Y. This is possible only if pebble P is bigger than pebble Q. Students choosing option B might be thinking that the amount of water in Jar Y is lesser, so more water needs to be displaced and hence Pebble Q is bigger. Students choosing option C might be thinking that the sizes of the pebbles are the same as the rise in water levels are the same. Students choosing option D might be thinking that the sizes of the pebbles can be compared only when all the other factors remain the same.

Papercode: 35224, Question no: 3					
Performance	Option A	Option B	Option C	Option D	No. of Students
Section	44%	9%	32%	12%	34
National	35%	11%	34%	20%	13500

Q 18 Diya wound a thick cotton thread around a wooden block as shown in Figure 1. She painted the block and the thread on it as shown in Figure 2.

Figure 1

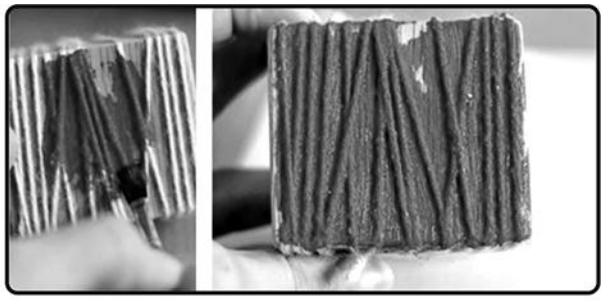


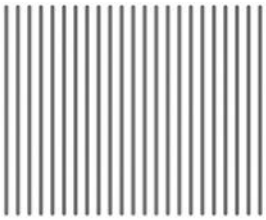
Figure 2

She GENTLY pressed the freshly painted block onto a white paper.

What will be printed on the paper?

A

B



C

D

This question tests the student's ability to predict a possible pattern based on analysis of the given information. Option D is correct because the thread wound around the wooden block will be raised above the surface of the block, and when the bock is pressed on to paper, only the thread will touch the paper and leave its imprint on the paper. Options A, B and C show a flawed understanding and interpretation of the given information. Options B and C show an incorrect visual interpretation of patterns.

Papercode: 35224, Question no: 18					
Performance	Option A	Option B	Option C	Option D	No. of Students
Section	32%	18%	6%	35%	34
National	34%	12%	6%	47%	13500

This section is crafted to aid students in practicing weaker skills. Use these in class discussions, gauge common understanding gaps, and tailor lessons. Answers and explanations are included. Employ a mix of question types to stimulate students' thinking. Here's how we've tailored the practice questions for each subject: For English, if reading comprehension is a challenge, we've provided full passages to work on. In Math, Science, and Social Studies, we've chosen questions based on areas where you've shown room for growth. Hindi practice focuses on non-reading comprehension skills. Computational Thinking covers a variety of skills across the class.

Skill: Extraction, translation and application of knowledge or information (43%)

Q 1

	Inhaled air	Exhaled air
Water vapour content	Less	More
Temperature	About room temperature	About body temperature
Carbon dioxide content	Less	More
Oxygen content	Less	More

The table above compares the composition of the air we breathe in (inhaled air) and the air we breathe out (exhaled air). One of the rows has incorrect information.
Which one is it?

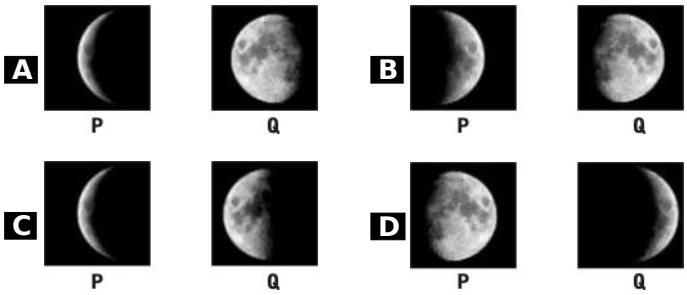
A Water vapour content
B Temperature
C Carbon dioxide content
D Oxygen content

Explanation:

D: We absorb oxygen from the air we breathe in for our bodies to use. We produce carbon dioxide in our bodies that is harmful to us and breathe it out. This is why the air we breathe out has less oxygen and more carbon dioxide.

Q 2 The moon changes its phase from new moon to full moon as shown here.

Which of the following match phases P and Q?



Explanation:

B: As the Moon's phases change from new Moon to full Moon there is a gradual increase in the side that is lit. This always remains on the same side until full Moon (check the option in which the dark region is decreasing on the same side of the moon - it is option B). A cannot be the answer because in A, the dark side is on the right whereas in the question, the dark side is on the left.

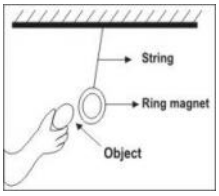
Q 3 Respiration is a process where oxygen is used, to release energy required for all the life processes and carbon dioxide is given out. Do plants respire?

A Yes, plants respire and it is called photosynthesis.
B No, plants do not respire for they give out oxygen.
C Yes, plants respire all the time because they need energy.
D Yes, plants respire but only during the night, not during the day.

Explanation:

C: Photosynthesis is the process by which plants make their food, which is stored in their bodies. Respiration is the process by which stored food is broken down for energy to carry out processes. Even plants need energy and hence, respiration does occur in plants - where oxygen is used up and carbon dioxide is given out. Remember that respiration and photosynthesis are two DIFFERENT processes!

Q 4 Geo: "Sparky, Look! I have hung my ring magnet. When I bring this object closer to the magnet, it moves towards my object.



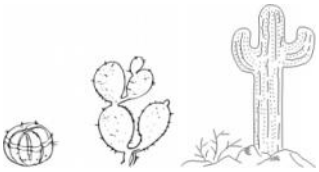
What do you think this object is made up of?"

A iron, because a magnet is attracted towards iron
B gold, because a magnet is attracted towards gold
C plastic, because a magnet is attracted towards plastic
D It could be any metal because a magnet is attracted towards all metals.

Explanation:

A: If you have played with magnets, you will know that they are attracted to some things, but not to everything. Magnets are not attracted to plastic and they are not attracted to all metals. Try taking a magnet close to aluminium or gold or silver. Hence, option D is incorrect.

Q 5 Shown here are some cactus plants. These plants grow in places where there is very little water. Most of these plants have no leaves.



Why are leaves absent in the cactus plant?

A The leaves dry up and disappear in the heat.
B The cactus does not need food, so leaves are not needed.
C Loss of water through leaves is minimised.
D The cactus looks beautiful if there are no leaves.

Explanation:

C: Plants lose a lot of water as water vapour through tiny openings in their leaves. Plants growing in very dry places often have either very small leaves or no leaves at all to reduce loss of water from leaves. All plants need food, so B cannot be the right answer.

Q 6 The view of the stadium shown here is from:

A inside the stadium on the ground level
B inside the stadium from one of the higher rows
C outside the stadium on the ground level
D outside the stadium, but from a great height

Explanation:

B: The view is from inside the stadium. If we were outside the stadium, the roof that we can see in the picture would prevent us from seeing what is below it. Since the figures on the field appear very small it must be from one of the higher rows.

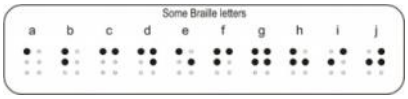
Q 7 Shikha, 12 yrs old, loved fruits and was wondering what would happen if she ate only fruits for a week. She chatted with her father, a dietician, who explained the reasons why it would NOT be a good idea. Which is the most important reason he must have given?

A You are a growing child; your body will not get enough proteins.
B This is the middle of summer and your body will not get the water it needs.
C You may fall ill easily because your body will not get enough vitamins and minerals.
D Your body will have problems getting rid of the undigested food because you will not get enough fibre.

Explanation:

A: Think of fruits like mangoes, bananas and juicy apples.They contain sugar and also vitamins and minerals but not enough proteins that you need for proper growth. Option D is not correct because there is plenty of fibre in whole fruits.

Q 8 Some people read from books using a special system of printing called Braille. The letters are made from patterns of little dots that rise above the surface of the paper. Which sense organ is involved when a person uses Braille?



A Eyes. **B** Ears **C** Skin **D** Nose

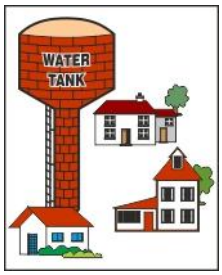
Explanation:

C: Braille letters are formed of patterns of dots that can be felt as bumps using your fingertips. The skin on your fingertips is responsible for your sense of touch that lets you feel with your fingertips.

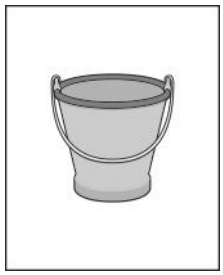
Skill: Knowledge of use of scientific instruments, tools and procedures (46%)

Q 1 Which of the following can hold ONLY about 20 litres of water?

A



B



C

D

Explanation:

B: Out of the 4 objects, only the bucket may have a capacity of 20 litres. Option A cannot be correct. A water tank would have a capacity many times greater than 20 litres. The other two objects are too small.

Q 2 Which thermometer is showing a reading closest to the temperature in your classroom on a hot summer day?

A

B

C

D

Explanation:

B: Hot summer temperatures are usually around 35 degrees C. The other options cannot be correct because they show temperatures that are too hot or too cold.

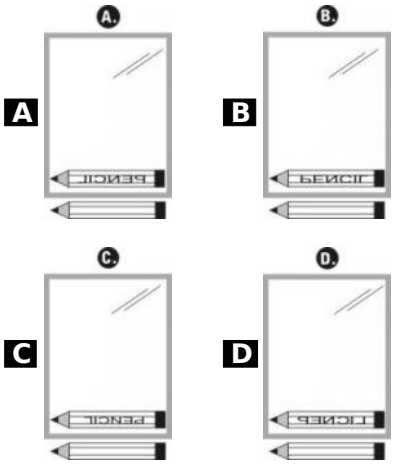
Q 3 Deepti was looking at the wire used to light a bulb. She saw that inside the plastic covering, there were thin metallic wires. How do the metal and the plastic help?

A The metal and the plastic together produce electricity.
B The metal does not bend and helps to keep the plastic stiff.
C The metal and the plastic allow electricity to flow through the wire easily.
D The metal allows electricity to travel easily and the plastic prevents shocks.

Explanation:

D: Metals allow electricity to flow through them and so are used in making wires. Plastic materials do not allow electricity to flow through them and are used to cover electric wires to prevent shocks.

Q 4 Geo had the pencil shown above. He put that in front of the mirror. Which of these could it look like?



Explanation:

A: Image of each letter formed in a mirror is reversed from left to right because when we keep the pencil in the front of the mirror; the letters are rotated from left to right (or vice versa) and not up to down.

Q 5 Many cycles have a bulb which glows when the cycle moves. There is no battery or cell. Where does the energy for the bulb to glow come from?

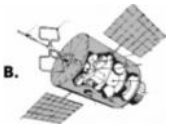
A (Energy is not required for the bulb to glow)
B From the sunlight falling on the bulb.
C From the pedalling action of the cyclist.
D The bulb is charged at home and so it glows.

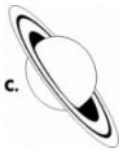
Explanation:

C: A device called a dynamo is used to transform the mechanical energy from the pedaling action of the cyclist to electrical energy that makes the bulb glow. It cannot be with the use of sunlight because as mentioned in the question, there is no cell or battery to make use of solar energy.

Q 6 One of these is the picture of an artificial SATELLITE. Identify it.

A A.

B B.


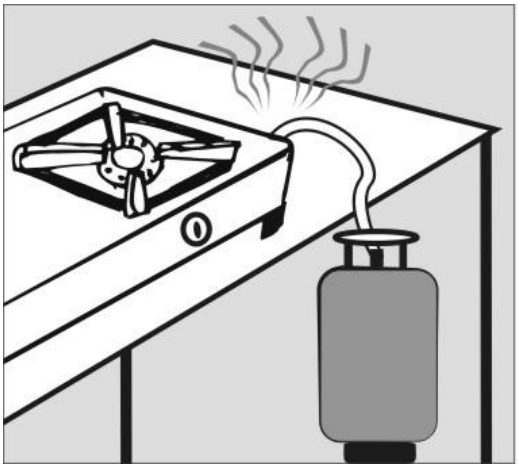
C

C.

D D.

Explanation:

B: The object shown in B is a satellite that man has sent into space. It orbits the Earth and is used for many things like weather forecasting and communication. A is not the correct answer. It shows the picture of a dish antenna that is used to send and receive signals to and from space (and satellites).

Q 7 When you smell a gas leak at home, you would take these immediate precautions to prevent a fire. Which of the following would be the order of the actions taken by you ?



P: *Call the gas agency for help*

Q: *Open all the doors and windows*

R: *Change the damaged rubber tubing*

S: *Switch off the gas regulator*

A R - P - Q - S **B** S - R - P -Q **C** S - Q - P - R **D** P - Q - R - S

Explanation:

C: Stopping any more gas from leaking is the first thing to do. Next is to reduce the amount of gas in the room by letting it out through open doors and windows. Then the other steps should be taken. Option B is not correct because the rubber tubing should be changed only by trained people from the gas agency.

Q 8 As Geo and Sparky were going back, Mr. Khoj told them, "Why don't both of you come to my lab tonight? I have got a new instrument that helps me check the patterns of rocks on the surface of the moon. Geo replied, "Sure sir. We will try and come tonight. It was a pleasure meeting you. Have a good day!"
As they were walking back, Geo asked Sparky, "Which instrument do you think Mr. Khoj was talking about?" Sparky correctly replied -

A a satellite **B** a magnifying lens **C** a powerful torch **D** a telescope

Explanation:

D: Telescopes can be used to get a magnified view of objects very far away from us like the moon. Even though satellites might be able to take pictures in the space, they function only when in the space and not when on the Earth’ surface.

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